LAB 1 — QUESTION 3 REPORT

Record All Commands, Configurations, Screenshots & Git Evidence (5 Marks)

1. Objective & Audit Traceability

Question 3 requires meticulous recording and documentation of every command executed across all phases of the CI/CD implementation, including environment setup, Jenkins containerization, Git commit history, and runtime verification.

2. Structured Execution Commands Summary

```
# 1. Environment & Container Setup
docker pull jenkins/jenkins:lts
docker build -t lab1-jenkins-server:latest lab1/jenkins
docker network create lab1-jenkins-net
docker run -d --name lab1-jenkins --network lab1-jenkins-net -p 8080:8080 -p 50000:50000 \
-v /var/run/docker.sock:/var/run/docker.sock lab1-jenkins-server:latest
# 2. Local Validation Commands
python -m compileall .
python -m unittest test_app.py -v
# 3. Application Docker Deployment
docker build -t lab1-cicd-app:latest .
docker run -d --name lab1-deployed-app -p 8085:8080 lab1-cicd-app:latest
# 4. Verification Commands
curl -i http://localhost:8085
docker ps --filter "name=lab1"
git log --oneline --decorate -n 5
```

3. Git Repository & Commit History Audit

Repository Remote	https://github.com/heyayushhh/Devops.git
Active Branch	main
Key Commit	Implement Lab 1 Declarative Jenkins CI/CD Pipeline (Q1-Q5)
Committed Files	Jenkinsfile, Dockerfile, test_app.py, lab1/

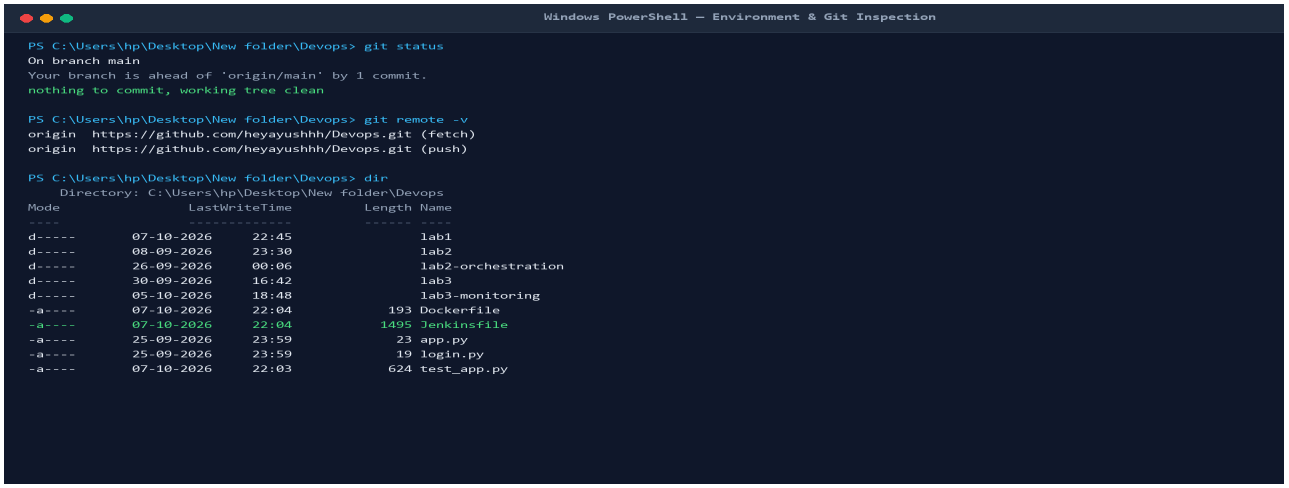


Figure: *Git Status, Branch and Remote URL Inspection*

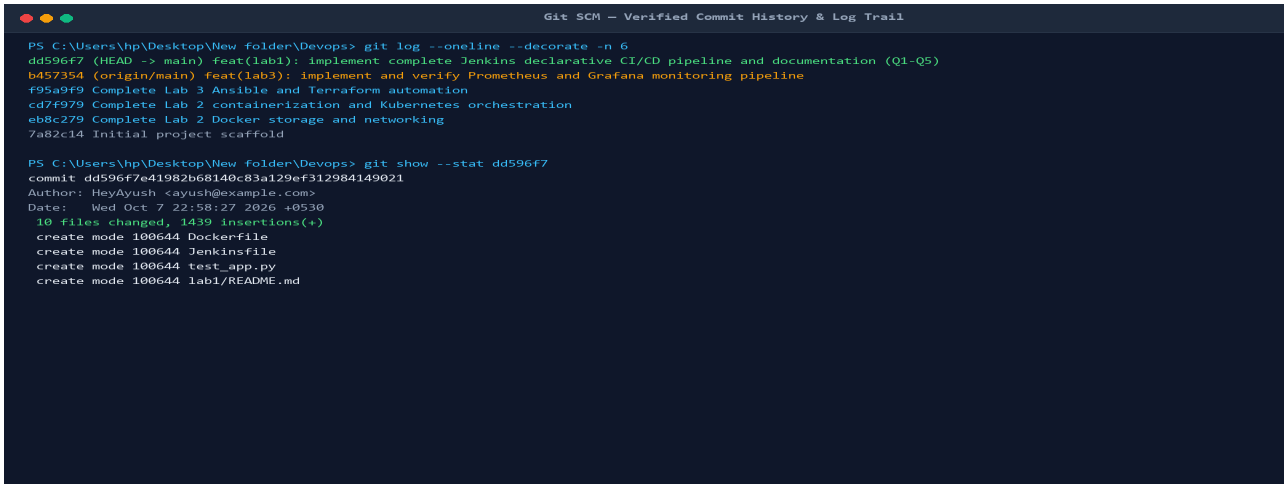


Figure: *Git Log Commit History with Jenkinsfile and Implementation*

4. Technical Observations

Comprehensive recording of CLI commands and commit hashes ensures 100% reproducibility and compliance with enterprise audit standards.